Laporan Sementara Elektronika Digital 2

Dasar dasar rangkaian sequensial 1

Anggota Kelompok :

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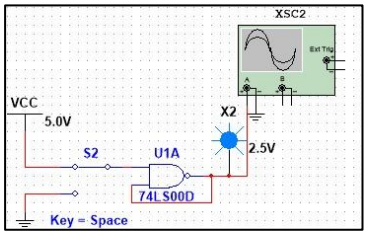
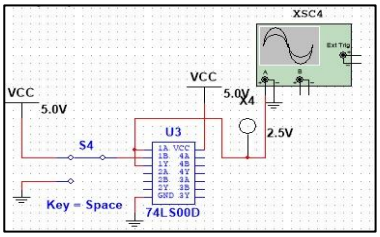
2. Muhammad Slamet Ridho Novianto (2425600050)
3. Arka Pandita Ari Dermawan (2425600057)

A. Percobaan Gerbang NAND

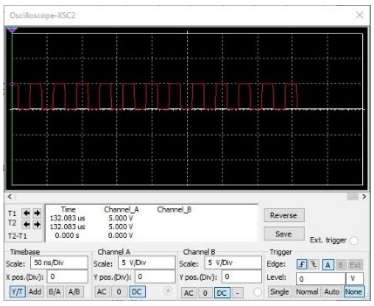
- Dengan umpan balik



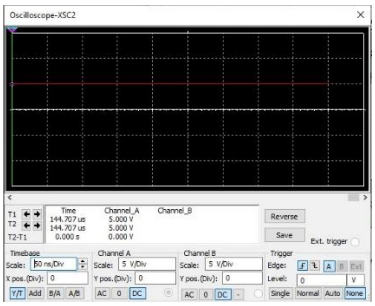
Rangkaian gate NAND



Rangkaian dengan IC



Oscilloscope logic “1”



Onscilloscope logic “0”

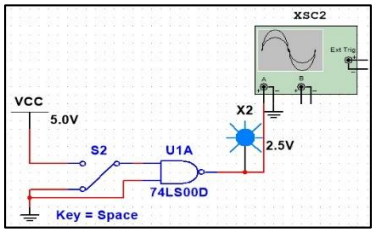
Kondisi Awal Output B =”0”

No	Present State (PS) Input A	Next State (NS) Output B
1	0	1
2	1	1

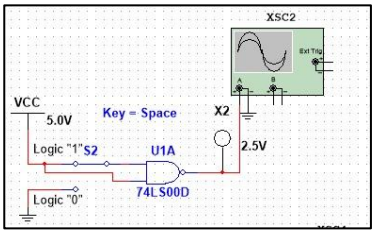
Kondisi Awal Output B =”1”

No	Present State (PS) Input A	Next State (NS) Output B
1	0	1
2	1	0

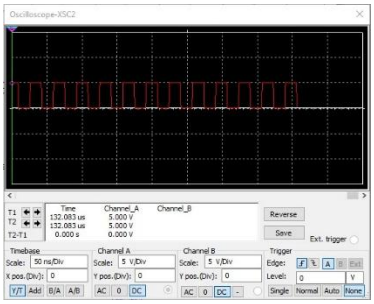
- Tanpa umpan balik



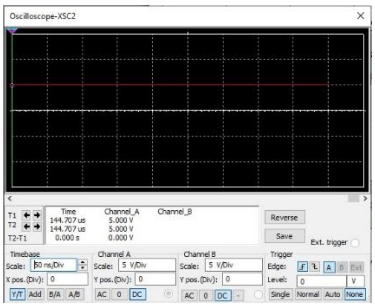
Kondisi salah satu input stabil “0”



Kondisi salah satu input stabil “1”



Oscilloscope logic “1”



Onscilloscope logic “0”

Kondisi salah satu input stabil “0” Tanpa Umpan Balik

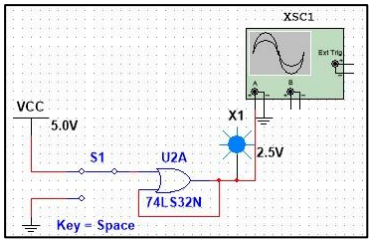
No	Present State (PS) Input A	Next State (NS) Output B
1	0	1
2	1	1

Kondisi salah satu input stabil “1” Tanpa Umpan Balik

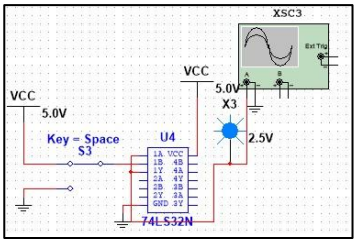
No	Present State (PS) Input A	Next State (NS) Output B
1	0	1
2	1	0

B. Percobaan Gerbang OR

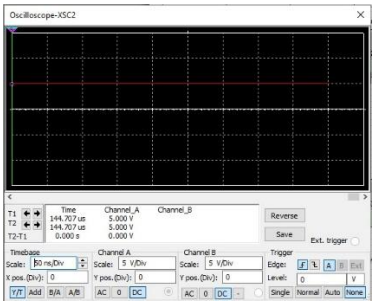
- Dengan umpan balik



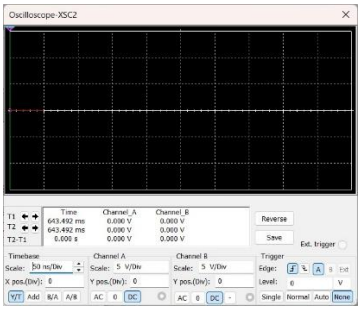
Rangkaian gate OR



Rangkaian OR dengan IC



Oscilloscope logic “1”



Oscilloscope logic “0”

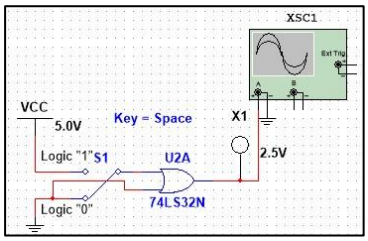
Kondisi Awal Output B =”0”

No	Present State (PS) Input A	Next State (NS) Output B
1	0	0
2	1	1

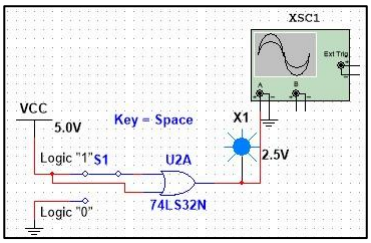
Kondisi Awal Output B =”1”

No	Present State (PS) Input A	Next State (NS) Output B
1	0	1
2	1	1

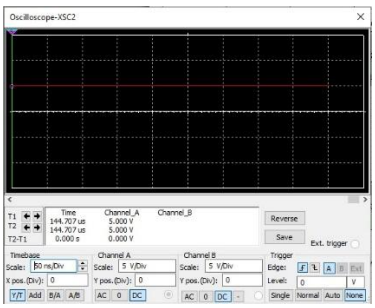
- Tanpa umpan balik



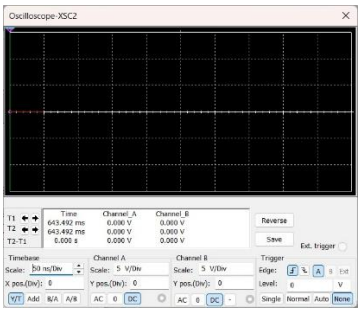
Kondisi salah satu input stabil “0”



Kondisi salah satu input stabil “1”



Oscilloscope logic “1”



Oscilloscope logic “0”

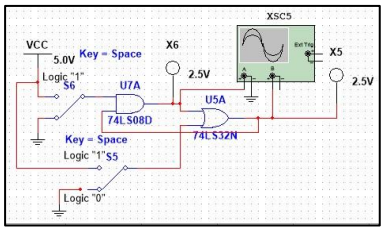
Kondisi salah satu input stabil “0” Tanpa Umpan Balik

No	Present State (PS) Input A	Next State (NS) Output B
1	0	0
2	1	1

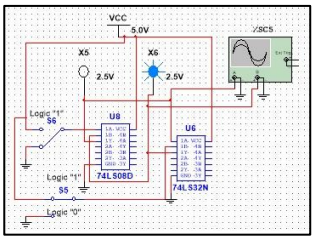
Kondisi salah satu input stabil “1” Tanpa Umpan Balik

No	Present State (PS) Input A	Next State (NS) Output B
1	0	1
2	1	1

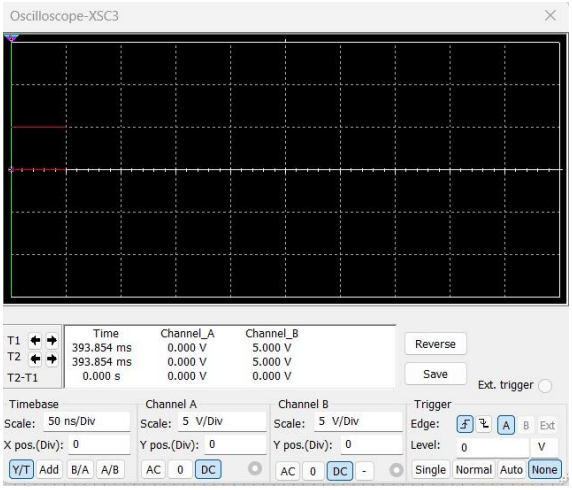
C. Percobaan 2 Jenis Gerbang (OR dan AND)



Rangkaian Gate



Rangkaian dengan IC



Hasil Oscilloscope

Dapatkan Tabel PS/ NS-nya dan state diagramnya

No	Present State (PS) Input A Input B	Next State (NS) Output X6 (X)	Next State (NS) Output X5 (Y)
1	00	0	1
2	01	0	1
3	11	1	1
4	10	1	1